# 2.3 introduction to memory

## What is memory?

Memory is learning that persists over time

Alzheimer's disease
- begins w/ difficulty remembering new info
- complex speech → simple sentences
- memory centers weaken & wither away
- sense of self ↓

Avg person stores & recognizes ~5000 faces

animals also can have strong facial-recognition

## Measuring Retention:

Evidence that learning persists incl. these retention measures:

- Recall - retrieving info thats not currently in conscious awareness
- Recognition - identifying previously learned items
- Relearning - learning something more quickly when u learn it again

Long after u cant recall, u may still be able to recognize

Recognition memory is very quick & vast

Our response speed recalling or recognizing info, & speed relearning, indicate memory strength.

Additional learning, esp. spread out over time, increases retention

Tests of recognition & time spent relearning demonstrate that:
we remember more than we can recall.

## What models do we use to understand memory?

Our information-processing model likens our memory to a computer:

- encode : Write information to the brain
- store : retain the information
- retrieve : Read, later, the information from the brain

But unlike a computer's sequential processing, our brain uses parallel processing:: processing many things simultaneously.

To focus on this process, connectionism, an info-processing model, views memories as products of interconnected neural networks.

Specific memories arise from particular activation patterns

You learn something new → ny connections change, neuroplasticity

To explain memory-forming, Richards Atkinson & Shiffrin proposed the three-staged multi-store model:

1) we record to-be-remembered info as a fleeting sensory memory

2) from there, process info → short-term memory, where we encode it thru rehearsal

3) finally, info → long-term memory, for later retrieval

# Working Memory

The Richards saw ST mem. as just a space for briefly storing recent thoughts & experiences.

Baddeley extended ST mem to:

Identified stage of mem ∴ working memory, where ST & LT memories combine

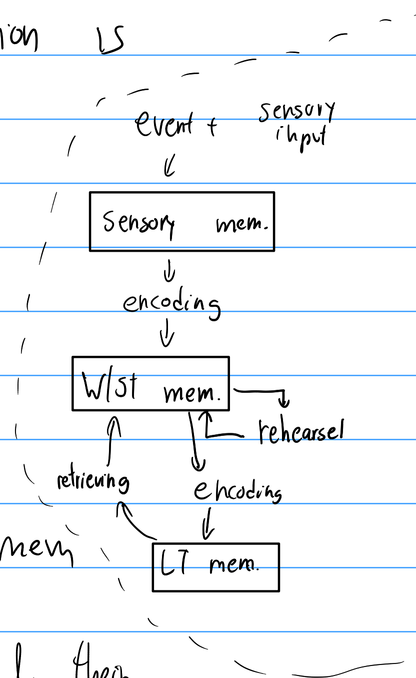
WM: 'scratch pad' where brain actively processes important info, by linking new exp. w/ LT memories.

WM helps us prolong memory storage:

through rehearsal over time ∴ maintenance rehearsal

through rehearsing info in ways that promote meaning, ∴ elaborative rehearsal.

As u integrate new info w/ existing LT mem, attention is focused.




This selective attention is coordinated by a Central Executive

w/o focused attention, info usually fades

phonological loop holds auditory info in ST mem

visuospatial sketchpad holds info about objects & their relation in space, in ST mem.

# How do Synapse-level changes affect memory processing?

As u learn → flexible brain is changing.

increased activities in particular pathways → nu interconnections are forming & strengthening.

neurogenesis, formation (genesis) of neurons, is happening

Experience & learning can increase, even 2x, # of synapses

rapidly stimulating certain memory-circuit connections
increased sensitivity for hours to even weeks.

Sending neuron needs less prompting to release its NuTr
& ∃ more connections

this increased efficiency of potential nu firing :: Long-term potentiation.

LTP provides nu basis for learning & remembering associations

potentiate something :: increase power or likelihood
∴ long-term potentiation :: long lasting increase in cell's firing power.

LTP is a physical basis for mem.

block LTP -> learning ↓

After LTP, old memories will be fine, but an electric current can hurt very recent memories.

which is why being knocked unconscious is not remembered
Wmem had no time to consolidate that info

One approach to mem-boosting drugs involves
NoTr glutamate, which is LTP-enhancing.

or drugs that boost the protein CERB, which boosts LTP process

Some mem-blocking drugs. can blunt intrusive memories.

Explicit memories :: facts, events, experiences, that we consciously perceive,

helped by frontal lobes & hippocampus.

Implicit memories :: learned skills & learned associations

helped by basal ganglia & cerebellum.

## Personal Connection

From my experience with computer science, the info-processing model is one I can really connect with, viewing the retrieval & encoding as read & write operations.

## MCQ

1. d  2. b  3. c  4. c  5. c  6. c

d?